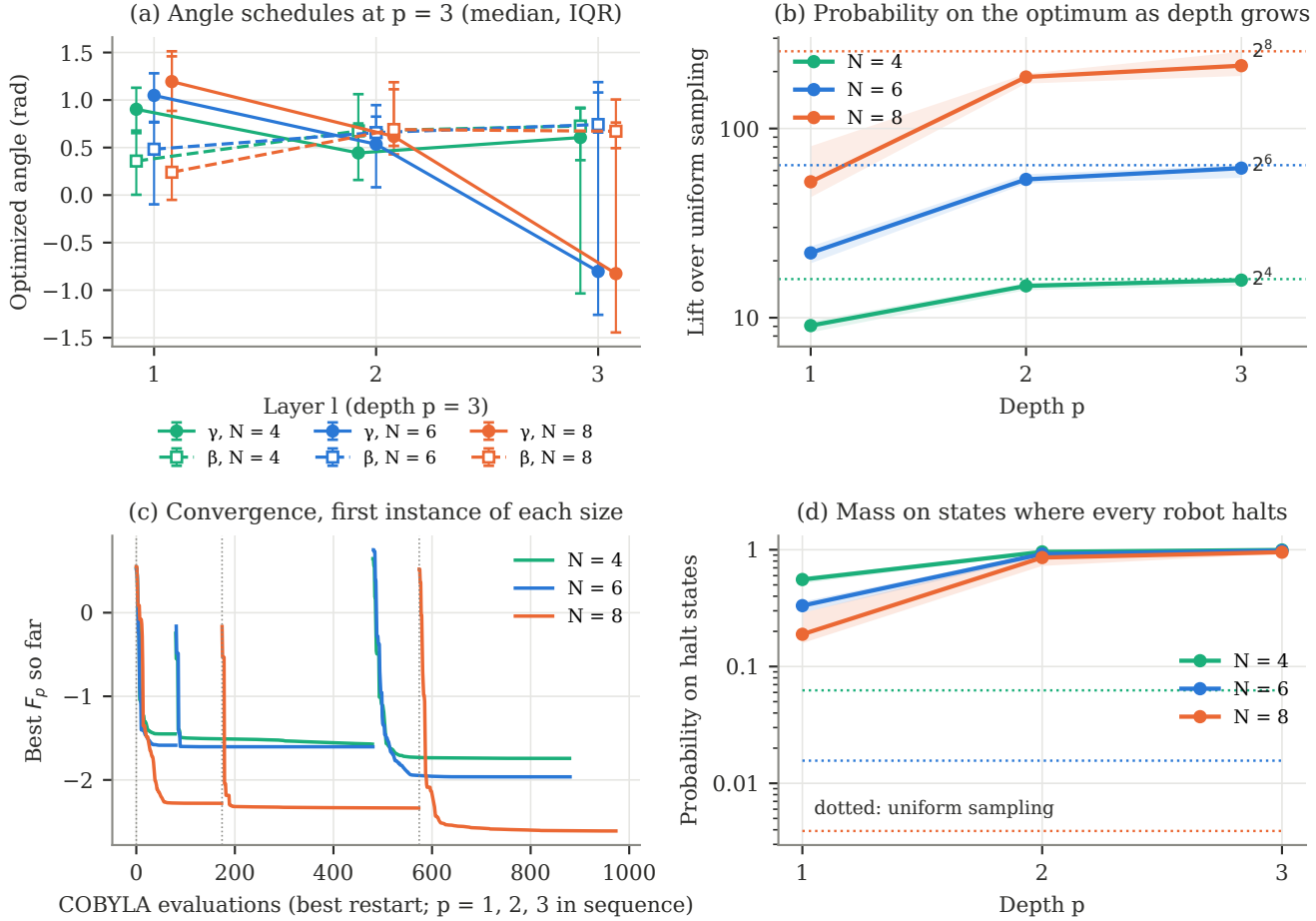
(a) Angle schedules at p = 3 (median, IQR)
Optimized angle (rad)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
1
2
3
Layer l (depth p = 3)
γ, N = 4
β, N = 4
γ, N = 6
β, N = 6
γ, N = 8
β, N = 8
(b) Probability on the optimum as depth grows
Lift over uniform sampling
100
10
N = 4
N = 6
N = 8
2^8
2^6
2^4
1
2
3
Depth p
(c) Convergence, first instance of each size
Best F_p so far
0
−1
−2
N = 4
N = 6
N = 8
0
200
400
600
800
1000
COBYLA evaluations (best restart; p = 1, 2, 3 in sequence)
(d) Mass on states where every robot halts
Probability on halt states
1
0.1
0.01
N = 4
N = 6
N = 8
dotted: uniform sampling
1
2
3
Depth p